

Wolf Vegas: 5-player scoring

How a 5-player Wolf Vegas hole is scored, why the rules are what they are, and the alternatives that were rejected and why. Shipped in v0.88.0 (2026-07-16).

Engine: `p4/components/golf/src/golf_scoring_wolf_vegas.c (settle_hole)`. Player count is 4 or 5 (`golf_wolf_vegas_player_count_ok`).

1. What changes at 5 players

Nothing about the Wolf side changes. The tee-order still rotates every hole and the player teeing last is still the Wolf. That already worked for any count, because `golf_wolf_vegas_for_hole` and `resolve_order` were written against `player_count` rather than a hard-coded 4.

What changes is the shape of the two sides:

Wolf's decision	4 players	5 players
Picks a partner	2 v 2	2 v 3
Lone Wolf (x2)	1 v 3	1 v 4
Blind Wolf (x3)	1 v 3	1 v 4

So 5-player introduces two genuinely new shapes: a three-man side, and a four-man opposing side.

2. Building each side's number

Rule: a side's Vegas number is its two lowest nets, low digit first.

This is not a new rule. The Lone Wolf branch has always built the three opponents' number from their best two of three. 5-player just applies the same idea to a side of any size:

- 2-man side: the two nets are the two lowest. Identical to classic Vegas.
- 3-man side: the two lowest of three. The third ball is ignored.
- 4-man side: the two lowest of four.
- 1-man side (Lone / Blind Wolf): his net doubled, e.g. 4 -> 44.

Why the two low balls and not, say, all three concatenated: a Vegas number has to be two digits for the difference to mean anything. Three digits versus two would make the comparison nonsense. Taking the two low balls keeps every side on the same scale no matter how many players are on it, and it matches how the group already thinks about Lone Wolf.

Nets are capped at 9 so the two-digit rule holds (a 10 would carry into a third digit).

The birdie flip

A side's number FLIPS (high digit first) when the **opposing** side birdied. Birdie means the side's lowest stroke is at least one under par, using gross strokes when the "Gross birdies flip" option is on, else net.

On a 3- or 4-man side, any member's birdie counts. A birdie is a birdie.

A flip is a no-op on a one-man side's doubled number, since 44 flipped is still 44. That is why an opponent birdie has no number effect against a Lone Wolf. It falls out of the math rather than being a special case.

3. Awarding the points

Rule: if the sides are equal size, each player settles the difference once. If they are unequal, each player settles the difference with each opposing player.

One rule covers every player count and every decision:

Shape	nw	no	Each wolf-side player	Each opponent
2 v 2 (4p, equal)	2	2	$+-diff$	$+-diff$
1 v 3 (4p, Lone)	1	3	$+-3diffmult$	$+-diff*mult$
2 v 3 (5p)	2	3	$+-3*diff$	$+-2*diff$
1 v 4 (5p, Lone)	1	4	$+-4diffmult$	$+-diff*mult$

The Lone/Blind multiplier and the per-hole hammer then scale both sides equally. The difference shown in the UI stays the base number difference, unscaled.

This rule was chosen because it reproduces the shipped 4-player behaviour exactly. The two existing branches were written years apart and use what look like different conventions (2v2 is flat, Lone is per-opponent), but both fall out of the single equal/unequal rule above. That mattered: 4-player rounds are what people actually play today, and the 5-player work rewrote the branch they run through.

4. Why not just "give all three players the difference"

The first instinct for 2v3 is that the three-man side each wins the difference, same as a 2v2 team each wins the difference. **This does not work, and the wager engine is where it shows up.**

Say the three-man side wins by 10.

- Give each of the 3 players +10. That is 30 points paid out.
- The wolf side only has 2 players to pay it. Flat -10 each is 20 points coming in.

30 out, 20 in. The table is short 10 points every hole. The points are not zero-sum, so `compute_settlement` produces per-player nets that do not cancel, and `golf_settlement_reduce()` emits a set of "X pays Y" rows that do not balance. At a \$0.25 per-point stake that is \$2.50 per hole conjured out of nothing. Nobody would notice on the leaderboard. Everybody would notice in the Money modal at the end.

Paying each of the 2 the full 30 does not work either, because then it is not "the difference" any more, it is 3x the difference on one side and 1x on the other with no principle behind the asymmetry.

The rejected middle option

Split it: the three-man side settles 30 against the wolf side as a unit, and the wolf side's 2 players split it, 15 each. That is zero-sum and fair. It is also fractional, and half-points in a game settled in quarters of a dollar is a bad time.

But notice what happens if you scale everything by 2 to clear the fraction: each of the 3 gets 20, each of the 2 gets 30. **That is exactly the per-opponent rule.** Two different derivations, same numbers. That is the main reason for confidence that the rule is right and not just convenient.

The consequence, accepted deliberately

The wolf side is outnumbered, so each wolf-side player has more at stake per hole than each of the three. That is real, and Ryan accepted it (2026-07-16) with the per-point stake left unchanged. It is the same "losses compound fast" property Lone Wolf has always had. The math is not hiding the skew of 5-player wolf, it is showing it.

5. Worked examples

All examples: par 4, no hammer unless stated, handicaps off (net = gross).

5.1 2 v 3, the ordinary case

Wolf A picks B. Opponents C, D, E.

```
Wolf side:  A=4, B=4          -> 44
Opp side:   C=5, D=5, E=6     -> two low 5,5 -> 55
diff = 44 - 55 = -11          -> wolf side is lower, wolf side wins, mag 11
nw=2, no=3, unequal          -> each wolf-side player +-3*mag, each opponent +-2*mag

A +33   B +33                (+66 total)
C -22   D -22   E -22        (-66 total)
```

E's 6 never mattered. The two low balls carried the side.

5.2 2 v 3 with a birdie flip

```
Wolf side:  A=3 (birdie), B=5 -> 35    (opponents did not birdie, no flip)
Opp side:   C=4, D=5, E=7    -> two low 4,5 -> would be 45,
                                   but the wolf side birdied -> FLIPS -> 54
diff = 35 - 54 = -19          -> wolf side wins, mag 19

A +57    B +57                (+114)
C -38    D -38    E -38      (-114)
```

The flip is what makes a birdie hurt. It swung the difference from 10 to 19.

5.3 1 v 4, Lone Wolf (x2)

```
Wolf:  A=4                -> 44 (his net doubled)
Opps:  B=4, C=5, D=5, E=6 -> two low 4,5 -> 45
diff = 44 - 45 = -1       -> wolf wins, mag 1
nw=1, no=4, unequal, mult=2 -> wolf  $+1*4*2 = 8$ , each opponent  $-1*1*2 = 2$ 

A +8
B -2  C -2  D -2  E -2    (-8)
```

A one-stroke difference became an 8-point swing. The wolf beat four players by a single stroke and collects from all four, doubled. This is the headline risk of 5-player Lone Wolf and it is working as designed.

5.4 1 v 4, Blind Wolf (x3)

Same as above but mult=3, so the wolf swings $\text{mag} * 4 * 3 = 12*\text{mag}$ and each opponent $3*\text{mag}$. Declared before anyone tees off.

5.5 Hammer stacking on 2 v 3

The hammer multiplies the hole's points and stacks on top of everything else.

```
2v3, mag 11, one hammer (x2):
  each wolf-side player:  $11 * 3 * 2 = 66$ 
  each opponent:         $11 * 2 * 2 = 44$ 
   $2*66 = 132 = 3*44$                 (still zero-sum)
```

Re-hammered to x8 on a Lone Wolf hole at x2, that is $\text{mag} * 4 * 2 * 8 = 64*\text{mag}$ for the wolf. Five-player Wolf Vegas can get expensive very fast. That is the group's problem, not the engine's.

5.6 Ties

Equal numbers means $\text{diff} = 0$ and nobody scores on the hole. No special handling.

5.7 Holes that do not settle

A hole scores only when **every** player has a net stroke AND the wolf has committed a decision ($\text{wolf_partner}[h] \neq -1$). Otherwise the hole contributes nothing and shows as unsettled. With 5 players that is five scores needed, not four.

6. Zero-sum, in general

For any shape, the wolf side has `nw` players each swinging `mag*no`, and the opposing side has `no` players each swinging `mag*nw`:

$$\text{nw} * (\text{mag} * \text{no}) == \text{no} * (\text{mag} * \text{nw})$$

Always balances. The equal-size case ($\text{nw} == \text{no}$) awards `mag` to each player on both sides, which balances trivially. Multipliers and hammers scale both sides by the same factor, so they cannot break it either.

This is worth preserving in any future change. The leaderboard tolerates an unbalanced points table silently; the Money modal does not.

7. What did not change

- **Wolf (the standalone game) is still 4-only.** Separate engine (`golf_scoring_wolf.c`), separate work.
- **No Supabase migration.** `wolf_order` is a JSON array that takes a fifth entry, `wolf_partners` and `wv_hammers` are keyed by hole, and scores are per-player rows. Nothing is player-count dependent.
- **The scorecard partner picker** already sized itself from `player_count`, so a fifth player's button appeared with no change.